

EVOLVINGWORLDNAV: PREDICTIVE 4D BELIEF FOR PERSISTENT NAVIGATION IN EVOLVING WORLDS

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ABSTRACT

Embodied navigation is typically formulated assuming that maps and memories describe a stationary world. This assumption breaks for long-lived agents, where environments evolve during periods of absence and historical observations can become correct yet decisionally stale. We formulate *persistent navigation in evolving worlds*, where an agent must infer the current task-relevant state from incomplete temporal experience, act under uncertainty, and update its beliefs through physical evidence. We introduce EVOLVINGNAV, a predictive embodied-agent framework centered on P4D-NAV, which transforms evidence-grounded 4D memory into an actionable belief over the present world. A continuous-time temporal model captures irregular histories, while a persistence–relocation formulation predicts whether previous states remain valid or where objects are likely to transition. The resulting belief guides cost-aware navigation and is refined through visibility-aware Bayesian updates during interaction. We further introduce P4D-BENCH, a 54-scene benchmark with **803.68k** tasks spanning state prediction, object navigation, vision-and-language navigation, and embodied question answering. Across diverse benchmarks, EvolvingWorldNav achieves **63.22%** HL-SR and **48.83%** HL-SPL on FindingDory, **35.43%** SR on GOAT-Bench, and **86.18%** Search SR with **70.15%** SPL on P4D-Bench. Controlled static, routine, and random worlds further show that the agent learns temporal evolution rather than exploiting scene-frequency shortcuts. Our work shifts embodied navigation from acting on remembered past observations toward maintaining, testing, and acting on beliefs about the world as it is now.

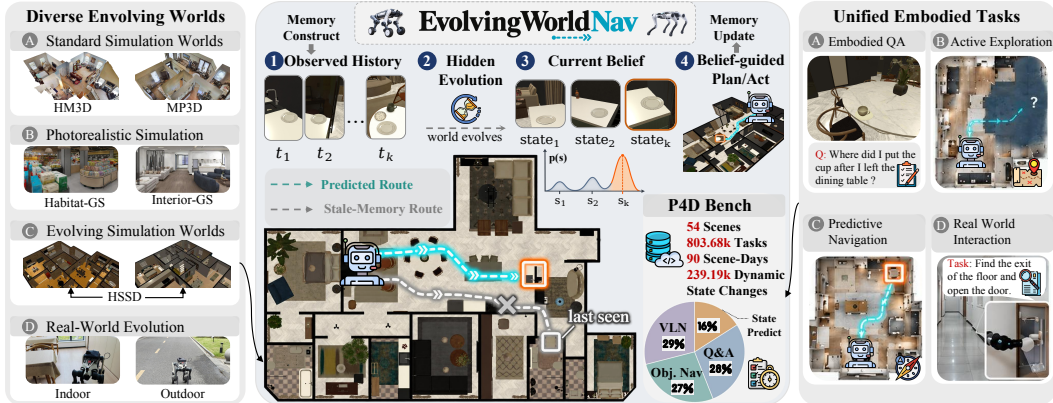


Figure 1: Overview of EVOLVINGNAV for predictive embodied reasoning and action in evolving worlds.

1 INTRODUCTION

Embodied navigation requires an agent to ground a goal, maintain a world model, and select actions from egocentric observations. Most formulations nevertheless adopt a snapshot abstraction: the map or memory is assumed to describe the same world state in which navigation is executed (Chaplot et al., 2020; Huang et al., 2023; Gu et al., 2024). Long-lived agents violate this assumption. Homes, offices, and campuses evolve while unobserved as objects are relocated, spaces are rearranged,

and accessibility changes. The agent may therefore retain evidence that is historically correct yet decisionally stale, pointing to an obsolete goal or an inefficient route.

Long-term deployment creates a dual tension. A persistent agent is not a persistent observer: as its *4D history* grows, individual records become stale, yet repeated experience reveals environmental dynamics, routines, and personal habits. More history therefore does not automatically mean better knowledge of the present. Meanwhile, history grows without bound, whereas decision-time context and computation must remain controlled. Most embodied memories are retrospective, organizing or retrieving what was observed (Liu et al., 2024; Kurenkov et al., 2023); a predictive memory should instead infer *what task-relevant state is most likely true now, how uncertain that belief is, and how navigation should adapt*. We seek a decision-preserving interface in which the episodic store retains grounded facts while the action policy consumes a compact, task-conditioned belief that preserves prediction and search utility.

Recent work models temporal persistence, future relations, household routines, and targets that move during navigation (Saavedra-Ruiz et al., 2026; Patel & Chernova, 2023; Argenziano et al., 2026; Dorbala et al., 2026). These approaches establish that dynamics matter, but leave a distinct problem insufficiently specified: navigation through changes that occur *between* observations. In this regime, neither a freshly updated map nor the trajectory of a currently observed target is available. The agent must infer the latent present from its own incomplete past and decide which physical action will most efficiently confirm or refute that inference.

Positioning against prior work. Our formulation separates four sources of difficulty that are often studied independently. Conventional ObjectNav and VLN primarily reason over an episode-level snapshot (Anderson et al., 2018; Savva et al., 2019; Chaplot et al., 2020); lifelong and dynamic mapping revise memory after a change is observed (Narayana et al., 2020; Liu et al., 2024); predictive scene models forecast future relations or placements, commonly as an open-loop prediction or ranked hypothesis set (Saavedra-Ruiz et al., 2026; Argenziano et al., 2026); and moving-target navigation plans against motion during execution (Dorbala et al., 2026). We instead study an agent that returns after an observation gap, faces a world whose task-relevant state may have changed unseen, and must close the loop between a calibrated current-state belief and evidence-seeking navigation. The key object is therefore not a more recent map or a single predicted destination, but an *action-facing belief over the present* that is repeatedly tested and revised. This positioning defines persistent navigation in evolving worlds as a problem at the intersection of long-term memory, temporal prediction, active perception, and embodied planning.

As illustrated in Figure 1, EVOLVINGNAV places this predictive loop in a unified embodied system. It constructs memory from conventional simulation (HM3D and MP3D), high-fidelity simulation (Habitat-GS and Interior-GS), dynamic HSSD scenes, and indoor or outdoor real-world observations. When the world evolves out of view, the agent transforms observed history into a belief over the current state, uses that belief to plan and act rather than blindly following a stale-memory route, and writes newly collected evidence back to memory. The same interface supports embodied question answering, active exploration, predictive navigation, and real-world interaction. P4D-BENCH provides the controlled simulated evaluation, while the real-world panel illustrates the deployment interface rather than a benchmark result.

We formulate this broader setting as **persistent navigation in evolving worlds**: before a query, an agent has accumulated timestamped observations through intermittent visits, while the task-relevant world state may have changed during the resulting observation gaps. We operationalize its central uncertainty through **persistent dynamic object navigation**, where a portable target may undergo hidden transitions after its last positive observation. This controlled instantiation is deliberately narrower than the new problem: it makes the latent state, evidence, and navigation cost measurable without reducing the conceptual setting to object displacement alone. The agent receives neither the current target state nor the activity that caused the change and must navigate under a fixed budget using only causal memory, candidate receptacles and viewpoints, and online RGB-D observations. Our main protocol freezes the target after the query to isolate reasoning over stale history; an extension allows another hidden transition during navigation.

We introduce EVOLVINGNAV, a long-term embodied agent centered on **4D Predictive Memory**, and instantiate its predictive-navigation core as P4D-NAV. Its 4D store maintains persistent identities, time-valid states, and observation provenance in a common frame. A causal retriever selects target-relevant history, and a lightweight continuous-time Transformer represents irregular gaps and calendar

periodicity. A structured head factorizes the current belief into the probability that the last-seen state persists and a pointer distribution over relocation candidates. This matches the navigator’s physical alternatives, supports variable candidate sets, and exposes the belief—rather than the event sequence—as the memory interface to the action policy.

Prediction is embedded in a closed decision loop. Concentrated beliefs support a direct attempt; diffuse beliefs induce probability–cost-aware exploration. Navigation returns new RGB-D evidence, and a visibility-aware Bayesian update suppresses an inspected hypothesis only when the target should have been observable there. The agent then replans from the posterior. The same controller handles rare, routine, personal, and irregular changes without an oracle dynamics label, connecting history $\rightarrow$ current-state belief $\rightarrow$ plan/action $\rightarrow$ evidence $\rightarrow$ memory update.

We evaluate this capability with P4D-BENCH, a 54-scene benchmark containing 803.68k task instances across state prediction, object navigation, vision-and-language navigation, and embodied question answering. Its evolving HSSD histories combine longitudinal activity records, real human–object interaction traces, and household state-transition programs. Our controlled analysis measures belief quality, navigation efficiency, evidence-aware recovery, and generalization. Paired static, routine, and random worlds share the observed setup while differing only in the hidden transition process, testing whether gains arise from learnable temporal structure rather than scene frequency or route geometry.

Contributions. Our contributions are threefold:

- We identify and formulate **persistent navigation in evolving worlds**, a new embodied problem in which unobserved evolution makes long-term history simultaneously stale evidence about the past and predictive evidence about the present; persistent dynamic object navigation provides a controlled operationalization.
- We introduce EVOLVINGNAV and its P4D-NAV core with **4D Predictive Memory**, which transform causal history into an action-facing persistence–relocation belief and close the loop through physical evidence, memory update, and replanning.
- We introduce P4D-BENCH, a human-trace-grounded, 54-scene benchmark spanning state prediction, ObjectNav, VLN, and EQA with paired static/routine/random controls.

2 RELATED WORK

Embodied navigation and active search. Embodied navigation spans object-goal, language-conditioned, and exploratory tasks that couple semantic perception, spatial reasoning, and action (Anderson et al., 2018; Savva et al., 2019; Chaplot et al., 2020; Xin et al., 2026). Semantic maps and open-vocabulary scene representations improve generalization by associating language with metric or graph-structured space (Huang et al., 2023; Gu et al., 2024; Werby et al., 2024). Probabilistic object-search methods use object–room or object–object correlations to order exploration and update a belief after observations (Zheng et al., 2022; Ginting et al., 2024). These formulations generally assume that the task-relevant world state remains consistent with the map during an episode. P4D-NAV addresses a different source of uncertainty: historically valid observations can become decisionally stale because the environment evolves between visits.

Training-free agents increasingly compose frozen foundation models, maps, and navigation primitives, whereas trained systems learn navigation actions from task trajectories or benchmark-specific adaptation (Yokoyama et al., 2024a; Chen et al., 2025; Ziliotto et al., 2025; Zhu et al., 2025; Wu et al., 2026; Zhang et al., 2025; Wang et al., 2026b). We evaluate both regimes rather than comparing only against methods with the same training recipe. The cross-environment study further separates standard mesh simulation, Gaussian-splatting simulation, and evolving worlds, using a canonical task for each environment so that visual fidelity and hidden state change are measured without conflating their protocols.

Navigation in changing environments. Early work on dynamic mapping often filters moving content or updates a semantic map after changes are observed (Bescos et al., 2018; Narayana et al., 2020). Scene Graph Memory models a partially observable dynamic graph and predicts object–location links (Kurenkov et al., 2023), while DynaMem continually reconstructs an actionable spatio-semantic memory for mobile manipulation (Liu et al., 2024). Predictive spatio-temporal

scene graphs attach persistence and emergence models to open-vocabulary relations (Saavedra-Ruiz et al., 2026). FlowMaps learns multimodal continuous future object placements and uses ranked hypotheses for dynamic ObjectNav (Argenziano et al., 2026). Portable Object Navigation and Transit-Aware Planning model objects that move as the agent moves (Dorbala et al., 2026). Our setting is complementary: dynamics are inferred from the agent’s causal, intermittent history; object-specific regimes are hidden; and prediction is embedded in an observe–update–replan loop rather than consumed only as a fixed candidate ranking or known transition path.

Long-term embodied memory. Long-horizon benchmarks increasingly require agents to retain observations across exploration or interaction (Khanna et al., 2024b; Majumdar et al., 2024; Yadav et al., 2025; Zhang et al., 2026). GOAT-Bench evaluates multi-modal lifelong navigation in persistent but static scenes, whereas FindingDory evaluates recall and action over long interaction histories. Open-vocabulary 3D memories support spatial retrieval (Gu et al., 2024; Werby et al., 2024), while recent 4D representations index time-varying states or human-centered events (Sohn et al., 2025; Bartoli et al., 2026). These systems motivate persistent identity, temporal validity, and provenance. We use these properties as infrastructure, but focus the learning problem on a navigation-relevant belief about hidden evolution: whether an entity’s last state persists and, if not, which reachable state should be inspected next.

Human routines and personalized object dynamics. Household activities exhibit temporal and person-specific regularities. HOMER and SLATe-PRO predict routine object usage for proactive assistance (Patel & Chernova, 2023); STREAK studies continual adaptation under household context drift (Bartoli et al., 2024); and UCON evaluates navigation with a large user-habit knowledge base (Wang et al., 2026a). Existing dynamic-world generators commonly rely on hand-designed routines or language-model priors. P4D-BENCH instead normalizes complementary longitudinal activity, real object-interaction, and complete state-transition records into executable HSSD episodes. Its paired worlds preserve marginal scene and query factors while changing the transition mechanism, enabling controlled measurement of when routine learning helps navigation.

3 PREDICTIVE 4D BELIEF NAVIGATION

3.1 PERSISTENT NAVIGATION IN EVOLVING WORLDS

At the system level, the task-relevant state may include object placement, spatial configuration, and accessibility. To isolate the coupling between hidden world evolution, predictive belief, and physical action, we formalize the core task through portable-object goals. Let an environment contain a navigable space $\mathcal{X}$, persistent entities $\mathcal{O}$, and candidate states $\mathcal{C}_o$ for each target $o \in \mathcal{O}$. A state identifies a semantically valid receptacle or region and its associated inspection viewpoints. Before a query at time t_q , the agent has collected only a partial history

$$\mathcal{H}_{\leq t_q} = \{(I_k, D_k, \xi_k, t_k) \mid t_k \leq t_q\}, \quad (1)$$

where I_k, D_k, ξ_k denote RGB, depth, and camera pose. The true query-time state $s_q^o \in \mathcal{C}_o \cup \{\text{unknown}\}$ and all transitions that occur outside the agent’s observations are hidden.

An episode supplies a target description, query time, initial pose x_0 , a causal memory materialized from $\mathcal{H}_{\leq t_q}$, and a finite action budget. At decision step j , the agent selects a candidate viewpoint or a low-level navigation action, receives observation z_j , and may terminate with STOP or NOT_FOUND. Success requires a correct stop near a valid goal viewpoint with the target visible. The policy should maximize success while minimizing travel and unnecessary inspections:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi}[\mathbb{I}(\text{success}) - \lambda_d L - \lambda_n N_{\text{inspect}}]. \quad (2)$$

The main protocol holds the target fixed after t_q , isolating whether stale history can be transformed into a useful belief about the present. An online-dynamic extension advances the world clock during execution and may apply an additional hidden transition outside the camera view.

Figure 2 presents the complete architecture. Timestamped RGB-D experience is consolidated into persistent entity histories, transformed into a task-conditioned predictive belief, and exposed to an embodied agent through typed memory and action skills. Physical evidence closes the loop through belief updating and replanning.

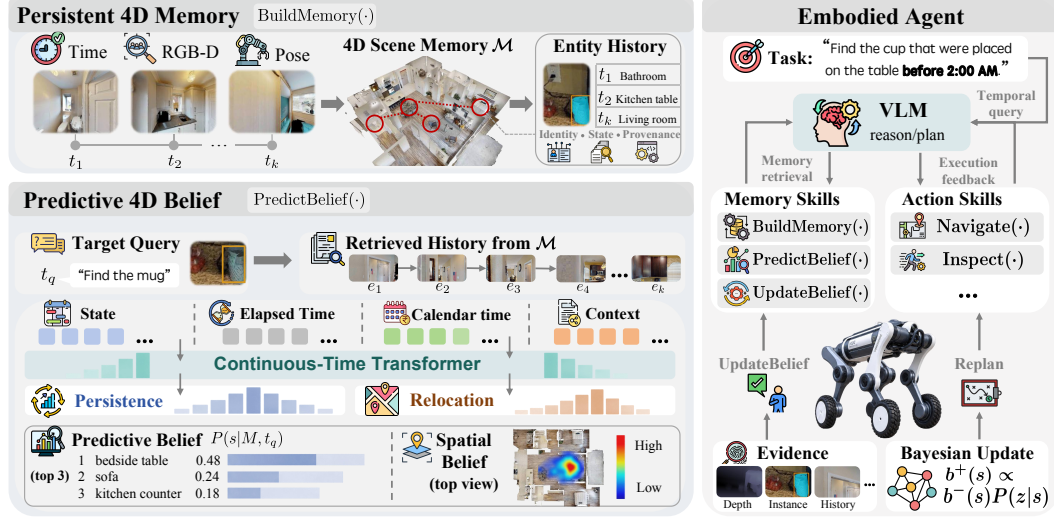


Figure 2: Architecture of EVOLVINGNAV, coupling persistent 4D memory, predictive 4D belief, and an evidence-updating embodied agent.

3.2 FROM CAUSAL 4D MEMORY TO CURRENT-STATE BELIEF

Evidence-grounded 4D memory. The memory represents a persistent 3D semantic scene whose entity states evolve over time. Timestamped RGB-D observations are registered in a shared frame and associated across views using semantics, geometry, temporal consistency, and source confidence. Memory is materialized causally from versioned deltas:

$$\mathcal{M}_{\leq t} = \text{Materialize}(\Delta\mathcal{M}_0, \dots, \Delta\mathcal{M}_t) = (\mathcal{G}_t, \mathcal{V}_t, \mathcal{B}_t), \quad (3)$$

where $\mathcal{G}_t$ stores semantic–spatial relations, $\mathcal{V}_t$ stores time-valid entity versions, and $\mathcal{B}_t$ stores observation provenance. Persistent identity is separated from mutable state: a change appends a time-valid version without erasing earlier states, and every query-time view excludes evidence after t_q . Projection, association, and entity-version details are provided in the appendix.

Target-conditioned history and continuous time. For target o , a typed memory query resolves its persistent identity and returns the most recent K causal events

$$H_o = \{e_k = (o_k, s_k, t_k, z_k, c_k, v_k)\}_{k=1}^K, \quad (4)$$

where s_k is the observed candidate state; $z_k \in \{\text{seen}, \text{notseen}\}$ is the evidence type; c_k is perception confidence; and v_k measures whether that state was adequately visible. Context events may share an entity, receptacle, region, or time window with the target, but hidden activities and transition labels are never exposed. Each scene-specific candidate is encoded from its context, functional role, geometry, and instance identity; an additional `unknown` candidate absorbs states outside the mapped set.

An event token combines entity, state, evidence, confidence, and visibility with elapsed and calendar time:

$$\mathbf{x}_k = E_o(o_k) + E_s(s_k) + E_z(z_k) + E_c(c_k) + E_v(v_k) + \phi_\Delta(t_q - t_k) + \phi_{\text{cal}}(t_k), \quad (5)$$

where ϕ_Δ includes $\log(1 + t_q - t_k)$ and ϕ_{cal} contains sinusoidal hour-of-day and day-of-week features. A learned query token specifies the target, query time, and last positive state. A compact Transformer yields

$$(\mathbf{h}_1, \dots, \mathbf{h}_K, \mathbf{h}_q) = \text{CTTransformer}(\mathbf{x}_1, \dots, \mathbf{x}_K, \mathbf{x}_q). \quad (6)$$

It therefore models irregular gaps, repeated negative evidence, and event order without an oracle mobility type or the activity responsible for a hidden change.

Structured persistence–relocation belief. A direct classifier entangles whether a state changed with where it changed to. We factorize these questions. Let s_{last} be the last positive state. The persistence branch predicts

$$\rho_q = P(s_q^o = s_{\text{last}} \mid H_o, t_q) = \sigma(\mathbf{w}_\rho^\top \mathbf{h}_q). \quad (7)$$

For each candidate $i \neq s_{\text{last}}$, a shared pointer head computes

$$a_i = \frac{(W_q \mathbf{h}_q)^\top (W_c \mathbf{g}_i)}{\sqrt{d}} + \psi(\mathbf{h}_q, \mathbf{g}_i), \quad \pi_q(i) = \frac{\exp a_i}{\sum_{j \neq s_{\text{last}}} \exp a_j}, \quad (8)$$

where $\mathbf{g}_i$ is the candidate representation. The prior belief is

$$b_q^-(i) = \begin{cases} \rho_q, & i = s_{\text{last}}, \\ (1 - \rho_q)\pi_q(i), & i \neq s_{\text{last}}. \end{cases} \quad (9)$$

The model is trained end-to-end with $\mathcal{L}_{\text{state}} = -\log b_q^-(s_q^o)$. Thus the factorization is an inductive bias rather than an oracle change label: it can preserve a strong last-state hypothesis, concentrate on recurring transitions, or expose diffuse uncertainty to the downstream agent.

3.3 PREDICTIVE EMBODIED AGENT

EVOLVINGNAV treats the belief model as a tool inside an embodied decision loop, not as an open-loop destination predictor. Given a navigation or language goal, the agent (i) resolves the referenced entity and task constraints, (ii) queries the causal 4D memory, (iii) invokes P4D-NAV at the decision time, (iv) selects a navigation or inspection tool call from the returned belief and map costs, and (v) writes the resulting observation and posterior back to memory. A frozen high-level controller may express goals and tool calls in language, while the belief update and spatial interfaces remain typed and verifiable. The perception stack and low-level navigator are shared across all compared methods, so the learned component changes what the agent believes and where it gathers evidence, not its locomotion capability.

Belief-guided action. Each candidate i has reachable inspection viewpoints $\mathcal{W}_i$. Let $d(x, \mathcal{W}_i)$ be the shortest geodesic distance from the current pose and $\hat{v}_i$ the expected inspection coverage. The agent selects

$$i^* = \arg \max_{i \in \mathcal{C}_o \setminus \mathcal{I}} \frac{b(i)\hat{v}_i}{d(x, \mathcal{W}_i) + \lambda_{\text{inspect}}}, \quad (10)$$

where $\mathcal{I}$ contains sufficiently inspected hypotheses. A concentrated belief supports direct navigation; a diffuse belief induces cost-aware active inspection. The low-level navigator executes the chosen route and returns an RGB-D observation rather than a privileged success flag.

Evidence update and replanning. After inspecting candidate r , the agent obtains result z , detector reliability c , and effective visibility $v \in [0, 1]$. A positive instance match supports r ; a negative view should suppress it only when the target would have been observable:

$$P(z = \text{notseen} \mid s_q^o = i, c, v) = \begin{cases} 1 - cv, & i = r, \\ 1, & i \neq r. \end{cases} \quad (11)$$

The posterior is

$$b_q^+(i) = \frac{b_q^-(i)P(z \mid s_q^o = i, c, v)}{\sum_j b_q^-(j)P(z \mid s_q^o = j, c, v)}. \quad (12)$$

A clear empty view provides strong counter-evidence; an occluded view makes only a small change. The agent stores the inspection, visibility assessment, and posterior as a new causal memory event, removes only sufficiently tested hypotheses, and repeats the plan–act–observe cycle until it succeeds, exhausts the budget, or returns NOT_FOUND.

This single loop supports direct predictive navigation, belief-guided search, and recovery after an incorrect first destination. In online-dynamic episodes, the agent predicts at the candidate arrival time rather than only at departure:

$$b_{t+\hat{\tau}_i}^- = F_\theta(H_o, t + \hat{\tau}_i), \quad (13)$$

and replans after every action chunk. No explicit router chooses a “static” or “dynamic” policy; behavior emerges from the current belief, travel cost, and newly acquired evidence.

4 P4D-BENCH: EVALUATING NAVIGATION IN EVOLVING WORLDS

Existing navigation benchmarks typically expose a fixed scene or reset the environment between episodes, making the agent’s latest map a valid account of the world. They therefore cannot isolate the central difficulty of our setting: the world may change during an observation gap, while only the agent’s causal history is available at decision time. We introduce P4D-BENCH as a dedicated benchmark for this gap. The full release contains 54 scenes and 803.68k task instances over longitudinal hidden state changes. It couples continuous world evolution with executable embodied episodes so that the same latent state can be evaluated through prediction, navigation, language-conditioned behavior, and embodied question answering.

4.1 DESIGN PRINCIPLES AND SCOPE

P4D-BENCH is designed around four principles. First, *temporal continuity*: examples are sampled from persistent scene histories rather than independent object placements. Second, *causal observability*: the public memory contains only evidence that the robot could have acquired before the query. Third, *physical executability*: predicted states correspond to collision-free placements with reachable, visibility-checked inspection viewpoints. Fourth, *controlled dynamics*: paired worlds keep the scene, target, query, and marginal placement factors fixed while varying the hidden transition mechanism. These controls separate predictive reasoning from scene frequency, route geometry, and privileged simulator access.

The task pool comprises current-state prediction (16%), object navigation (27%), vision-and-language navigation (29%), and embodied question answering (28%). Prediction probes the belief independently of control. Navigation tests whether that belief produces useful destinations, search orders, and recovery. VLN supplies language-conditioned goals over the same evolving histories, while EQA probes whether memory preserves the evidence needed to reason about current and historical states. The controlled experiments in this paper emphasize state prediction and navigation, with EQA as a memory diagnostic; the remaining task annotations are included to support broader study of evolving-world agents.

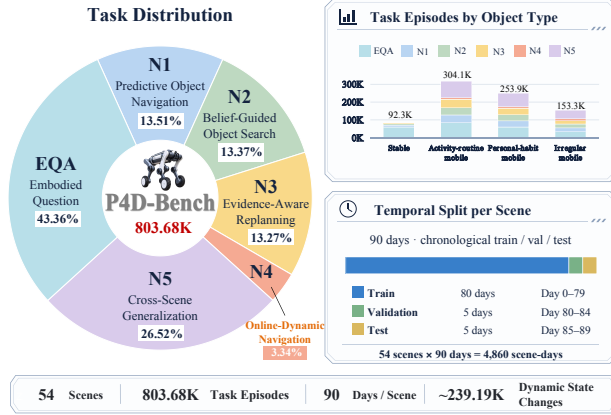


Figure 3: Overview of P4D-BENCH evaluation protocols.

4.2 HUMAN-TRACE-GROUNDED WORLD EVOLUTION

Source alignment. We derive complementary statistics from longitudinal smart-home records such as CASAS (Cook et al., 2013) and ARAS, and action-object evidence from EPIC-KITCHENS-100 and OPPORTUNITY. These sources are normalized into a common event schema linking time, activity, actor context, object category, source state, destination state, and evidence provenance. They constrain schedules, activity-object associations, transition timing, and placement tendencies; no source trajectory is copied verbatim into simulation. Unsupported quantities are explicitly marked as benchmark-authored rather than presented as measured human behavior.

Persistent household generation. Across 54 HSSD scenes (Khanna et al., 2024a) and multi-day household histories, we freeze household-level factors including resident schedules, room preferences, orderliness, object ownership, and placement habits. We then jointly generate activities, resident region trajectories, and object transitions in chronological order. Frozen factors make earlier observations informative about later states, whereas stochastic events, delayed returns, and exceptions prevent the histories from becoming deterministic scripts. The audited configuration yields 42.6 region transitions per resident-day, 6.56 relocations per personal object-day, 52.6% within-region relocations, and zero terminal collisions.

4.3 CONTROLLED DYNAMICS AND MOBILITY REGIMES

Every relocation is attributable to an activity, resident transition, object lifecycle event, or tidying event. Targets follow one of four complementary mechanisms shown in Table 1. Stable or rarely moved objects measure whether a predictive method preserves valid history. Routine objects follow repeated activity chains. Personal objects alternate between placed and carried states and exhibit owner-specific return habits. Irregular objects use affordance-valid destinations without temporal or actor-specific predictability. Mobility labels support stratified analysis but are hidden from the agent.

Table 1: Mobility regimes in P4D-BENCH. Labels are used only for stratified evaluation and are not provided to the agent.

Regime	Mechanism	Frozen signal	Role
Stable/rare	Infrequent events	Home/return tendency	Persistence control
Routine	Activity state machine	Chain-specific locations	Routine prediction
Personal	Placed/carried lifecycle	Owner placement habit	Long-term memory
Irregular	Affordance sampling	Affordance only	Uncertainty control

For each compatible history and query, we construct paired *static*, *routine*, and *random* worlds. Static preserves the last supported state; routine samples from the household’s frozen transition process; and random matches legal candidates and marginal movement statistics while breaking the learnable temporal dependency. Pair members share all public setup variables. Consequently, improvement specific to routine worlds is evidence that a model uses historical regularity rather than category frequency or spatial convenience. Audits additionally verify that personal objects do not simply track their owners and that actor or activity identity does not predict irregular destinations beyond affordance.

4.4 PHYSICAL REALIZATION AND CAUSAL OBSERVATION

Generated transitions are replayed chronologically in Habitat. Only semantically valid, collision-free placements with at least one reachable inspection viewpoint enter the benchmark. For every candidate, we record navigability, geodesic cost, camera frustum coverage, unoccluded target fraction, and success viewpoints. These checks prevent a method from being penalized for an impossible placement or an unreachable observation.

A robot patrols using the map, time, travel cost, and its own accumulated history; it never observes current object states, resident locations, hidden activities, or generator latents. Region visits produce local RGB-D snapshots that are converted into positive and negative observations with visibility metadata. For a target queried at t_q , the public input terminates at or before t_q and contains the observed history, last supported state, elapsed time, calendar context, legal candidates, and spatial context. The evaluator retains the current state, hidden transitions, query-time activity, mobility regime, and oracle viewpoints. Query times cover both the natural temporal distribution and mechanism-relevant windows, such as stages of a routine or the interval after a personal object is dropped.

4.5 TASKS AND EVALUATION PROTOCOLS

P4D-BENCH derives all prediction and navigation episodes from the same target–time queries (Figure 3 and Table 2). The prediction track asks for a calibrated belief over the current region or receptacle without navigation confounds. A navigation episode augments that query with a start pose, candidate inspection viewpoints, a path budget, and frozen success conditions. N1 permits only the Top-1 predicted destination and measures whether prediction becomes a useful physical decision. N2 exposes the full belief and measures cost-aware search. N3 updates the belief after visibility-qualified observations and tests recovery from an incorrect first destination. N4 advances the world clock during execution and can introduce another hidden transition. N5 is not a separate task; it applies the preceding protocols to held-out scenes and households.

The EQA suite queries existence, location, temporal order, activity association, count, comparison, and compositional relations over the same causal memories. Answers are evaluated against a normalized ontology, and questions requiring future evidence are excluded. VLN episodes replace an object label with a language-conditioned goal whose referents and constraints are grounded in the same versioned scene history. Across tracks, methods receive identical public observations,

Table 2: Prediction and navigation protocols in P4D-BENCH.

Task	Decision signal	Primary measure
P Current-state prediction	Region/receptacle belief	$R@k$ / NLL
N1 Predictive navigation	Top-1 belief	First-Go SR
N2 Belief-guided search	Full belief + cost	Search SR / cost
N3 Evidence-aware replanning	Updated posterior	Recovery SR
N4 Online-dynamic navigation	Arrival-time belief	Online Recovery SR
N5 Cross-scene generalization	Any protocol above	Held-out SR

candidates, geometry, starts, controllers, and budgets; only their memory representation, current-state belief, and resulting decisions differ.

4.6 SPLITS, LEAKAGE PREVENTION, AND QUALITY CONTROL

We split complete temporal sequences rather than individual queries so that no future portion of a household day can leak into an earlier decision. Held-out homes are reserved for cross-scene evaluation, and paired static/routine/random members remain in the same split. Near-duplicate queries and transition groups are assigned atomically. The serialized release is audited for future timestamps, private event fields, answer-correlated filenames or indices, repeated event groups across splits, inconsistent paired variants, invalid physics, unreachable goals, and accidental dependence of the patrol policy on hidden state.

Results are reported by mobility regime, transition class, staleness, spatial difficulty, and generalization split. We also expose dataset cards and audit summaries for source mappings, authored assumptions, physical validity, candidate coverage, and leakage tests. Full schema definitions, episode validity rules, and metric implementations are provided in the supplement.

5 EXPERIMENTS

We evaluate three linked questions: whether P4D-NAV predicts the latent current state more accurately and with better calibration; whether its belief improves navigation destinations and search efficiency; and whether the closed-loop agent recovers when its initial prediction conflicts with physical evidence. Benchmark experiments follow the native protocols described below; within each comparison, methods share perception, candidate states, inspection viewpoints, low-level control, and action budgets.

5.1 EVALUATION MATRIX

Cross-benchmark evaluation. We evaluate the complete agent on P4D-BENCH, FindingDory (Yadav et al., 2025), and GOAT-Bench (Khanna et al., 2024b). The three benchmarks probe complementary capabilities rather than sharing a single pooled score. P4D-BENCH isolates belief over hidden current state and its effect on evidence-aware navigation. FindingDory evaluates long-horizon memory through high-level goal selection and conditional low-level execution; the compact comparison reports HL-SR and HL-SPL, while the appendix retains the full high- and low-level breakdown. GOAT-Bench evaluates lifelong multi-goal navigation; in addition to its official SR and SPL, we report Repeat-SR on subtasks whose target identity or containing region has appeared in the agent’s prior stream. The latter is an explicitly marked diagnostic of memory reuse, not an official leaderboard metric. Category, language, and image goals are disaggregated in the appendix. On P4D-BENCH, First-Go SR, Search SR, and SPL jointly expose prediction quality, eventual task success, and path efficiency. Every comparison uses the benchmark’s native protocol, and reported external numbers are separated from reproduced results.

Prediction-only study. We train on days 0–29, select configurations on days 30–34, and evaluate on days 35–39 of five HSSD scene histories (5,475/910/915 queries). Temporal evaluation pools the five scenes; leave-one-scene-out (LOSO) trains and selects on four scenes and tests on the fifth. All methods share query identities and legal candidates. We report temporal *changed* queries ($n = 322$), whose current state differs from the last positive observation, and pooled LOSO *change-balanced* queries ($n = 644$), with equal changed and unchanged counts within observation-age bins. These are conditional prediction metrics, not navigation success rates. The complete natural distribution

and implementation details appear in Appendix C.2. This single-seed study evaluates the full Ours-Memory configuration after development informed by earlier test results; its richer inputs and search budget differ from the frozen legacy baselines. A separate three-seed study with fresh synthetic trajectories and matched neural inputs tests whether its advantages survive stronger controls.

Cross-domain simulation. To distinguish temporal reasoning from renderer- or dataset-specific fitting, we evaluate one canonical task in each environment: R2R-CE on MP3D, HM3D-OVON on HM3D, the native PointNav protocol on Habitat-GS, SAGE-Bench on InteriorGS, and P4D-BENCH on dynamic HSSD scenes. Each column keeps its official split, goal specification, and success rule; scores are not pooled across tasks. We hold perception and low-level control fixed within a column and report either exact-protocol published results or matched runs from released checkpoints. This design covers standard meshes, Gaussian-splatting simulators, and evolving worlds without treating heterogeneous tasks as a single metric.

5.2 REAL-WORLD EVALUATION

We further evaluate EVOLVINGNAV on three commercial legged robotic platforms, DEEP Robotics LYNX M20, X30, and Lite3, across both indoor and outdoor environments. M20 serves as the primary platform for quantitative evaluation, while X30 and Lite3 are used to examine cross-platform transfer. The memory representation, prediction module, belief update, and high-level search policy are shared across platforms; only the robot-specific perception and low-level navigation interfaces differ. Manufacturer-reported platform specifications are provided in the appendix.

The real-world evaluation focuses on persistent target search under environmental changes that unfold across multiple observation sessions. In the indoor setting, both objects and people may change locations over time, with their histories exhibiting different degrees of temporal and activity-related regularity. The outdoor setting extends the same formulation to larger and less constrained spaces, where targets likewise undergo recurring or irregular location changes. In both settings, the agent receives historical observations before a query and must infer the target’s current likely state, prioritize candidate locations, navigate to gather physical evidence, and revise its belief when new observations contradict the initial prediction.

Targets remain stationary within an individual search episode, so the evaluation isolates hidden changes between observations rather than target motion during execution. Compared methods receive the same pre-query history, candidate locations, starting condition, perception interface, and platform-specific low-level navigation stack. Success requires autonomous online confirmation of the queried target within the search budget; missed detections, navigation failures, timeouts, and human interventions count as failures. We report Search SR, First-Go SR, travel distance, elapsed time, number of inspected locations, and recovery after an incorrect first prediction. Detailed task construction, robot specifications, and scoring procedures are provided in Appendix D.2.

5.3 BASELINES AND ABLATIONS

We select baselines from the implementations that have been validated in our evaluation stack and organize them by the capability they contribute. The *open-vocabulary and lifelong navigation* group contains VLFM (Yokoyama et al., 2024a), ZSON (Majumdar et al., 2022), the FindingDory agent (Yadav et al., 2025), and LagMemo GLUE (Zhou et al., 2025). These methods test frontier-based semantic search, learned zero-shot navigation, long-horizon episodic recall, and persistent 3DGS language memory, respectively. The *structured spatio-temporal memory* group contains HOV-SG (Werby et al., 2024) and DynaMem (Liu et al., 2024), representing hierarchical open-vocabulary graphs and online dynamic metric-semantic memory. The *predictive world-model* group contains SLaTe-PRO (Patel et al., 2023), SGM+NEP (Kurenkov et al., 2023), FlowMaps (Argenziano et al., 2026), and the Perpetua predictor from PredictiveGraphs (Saavedra-Ruiz et al., 2026).

Navigation systems are executed through their released policies or benchmark adapters. Memory-only methods replace EVOLVINGNAV’s memory backend while using the same detector, goal interface, and low-level controller. Prediction-only methods receive the same causal history and candidate set; their state scores are passed to the same inspection planner. This component-aligned adaptation is reported separately from native published results and prevents a stronger detector or controller from being mistaken for better temporal reasoning. Last Seen, Frequency, Markov, and Random Forest

remain lightweight diagnostic baselines in the dedicated P4D-BENCH and predictor tables rather than the headline cross-benchmark comparison.

For the cross-environment evaluation in Table 4, baselines are additionally stratified by training regime. MTU3D, SoftNav, DUET, Uni-NaVid, NaVILA, and D3D-VLP represent navigation-trained or task-adapted policies (Zhu et al., 2025; Wu et al., 2026; Chen et al., 2022; Zhang et al., 2025; Cheng et al., 2025; Wang et al., 2026b). VLFM, AO-Planner, TANGO, MSGNav, and Uni-LaViRA represent training-free or target-benchmark zero-shot agents (Yokoyama et al., 2024a; Chen et al., 2025; Ziliotto et al., 2025; Huang et al., 2026; Ding et al., 2026). This selection spans classical policies and recent generalist agents while retaining only methods with reported scores on the selected task or sufficient released code and checkpoints for reproduction.

N4 additionally compares against Transit-Aware Planning under its required online target-path input (Dorbala et al., 2026). Oracle Current State provides the navigation upper bound. Oracle Activity supplies the ground-truth current activity and is reported only as a diagnostic rather than a fair baseline.

For EQA, all methods use the same question templates, answer ontology, and normalization. Queried information must be recovered from the memory available to each method. When a learned language readout is required, it is held fixed across memory methods so that the comparison primarily measures memory quality rather than language-model capacity.

The principal ablations remove continuous-time features; the persistence–relocation factorization; instance-specific history; calendar context; negative evidence; visibility calibration; Bayesian belief updating; or the full distribution in favor of Top-1 prediction. *P4D-Belief Prior* predicts once and searches without posterior updates, while full P4D-NAV closes the observe–update–replan loop.

We additionally replace the predictive component with Frequency, Markov, Random Forest, GRU, direct Transformer, and the structured continuous-time Transformer while keeping memory, candidates, and the agent policy fixed. To test whether the system depends on one proprietary controller, we run the same typed memory and navigation tools with GPT-family and Qwen-VL-family foundation models, plus a third VLM where available. Comparisons require fixed model versions, prompts, temperature, tool schemas, and inference budgets; these configuration fields belong to the reproducibility record in Appendix A.4.

5.4 METRICS AND REPORTING

We first assess predictive belief independently of navigation using state Top-1/Top- k accuracy, negative log-likelihood (NLL), Brier score, expected calibration error (ECE), true-state rank, mean reciprocal rank (MRR), and persistence AUROC.

For navigation, we report Success Rate (SR), SPL, path length, completion time, First-Go SR, number of inspected candidates, and search cost. N3 additionally reports Recovery SR after an incorrect first prediction and posterior true-state rank improvement after negative evidence. N4 reports Online Recovery SR, replanning count, replanning delay, and excess path length relative to a trajectory-aware oracle. This oracle is required because motion during execution makes the conventional shortest path to a fixed terminal goal ill-defined.

For EQA, we report answer accuracy by existence, location, temporal, activity, count, comparison, and reasoning question, together with category-macro accuracy. Structured outputs are normalized to the benchmark ontology before scoring.

Benchmark aggregate results include bootstrap confidence intervals over episodes and both micro and mobility-regime macro averages. We stratify results by paired world variant, mobility regime, transition class (*stay*, *same-region move*, *cross-region move*, *multi-move*, and *return*), staleness, spatial difficulty, grounding quality, and generalization split. The key causal contrast is the paired routine gain: improvement in the routine world relative to performance on its paired static and random variants. This contrast tests whether gains arise from persistent temporal regularities rather than scene geometry, identity, or generic search behavior. Implementation and reporting details appear in Appendices A.4 and E.1.

6 RESULTS AND ANALYSIS

The evaluation separates six questions: performance across embodied benchmarks, predictive navigation on P4D-BENCH, visual-domain generalization, the role of individual components, controller dependence, and indoor/outdoor object search. – denotes an unavailable measurement.

6.1 COMPARISON ACROSS EMBODIED BENCHMARKS

Table 3: Cross-benchmark comparison (%). EVOLVINGNAV reports mean $\pm$ standard deviation over five seeds.

Method	FindingDory		GOAT-Bench			P4D-BENCH		
	HL-SR $\uparrow$	HL-SPL $\uparrow$	SR $\uparrow$	SPL $\uparrow$	Repeat-SR $\uparrow$	First-Go SR $\uparrow$	Search SR $\uparrow$	SPL $\uparrow$
<i>Open-vocabulary and lifelong navigation</i>								
VLFM (Yokoyama et al., 2024a)	13.42	10.86	15.79	5.45	11.82	20.65	28.36	24.67
ZSON (Majumdar et al., 2022)	27.89	21.81	29.64	2.45	16.37	26.84	32.13	28.55
FindingDory Agent (Yadav et al., 2025)	52.44	–	–	–	–	–	–	–
LagMemo GLUE (Zhou et al., 2025)	32.79	23.25	20.61	1.74	11.27	30.18	42.27	33.82
<i>Structured spatio-temporal memory</i>								
HOV-SG (Werby et al., 2024)	21.47	15.20	28.37	2.23	12.38	36.46	52.32	40.61
DynaMem (Liu et al., 2024)	30.30	22.78	14.54	1.71	15.79	45.33	81.00	66.83
<i>Predictive world and state modeling</i>								
SLaTe-PRO (Patel et al., 2023)	18.18	14.53	9.41	0.99	6.38	13.00	36.33	26.37
SGM+NEP (Kurenkov et al., 2023)	34.20	19.48	10.67	1.64	2.56	27.26	41.82	37.00
FlowMaps (Argenziano et al., 2026)	3.03	3.03	15.07	0.32	6.25	4.33	22.67	14.35
Perpetua (Saavedra-Ruiz et al., 2026)	24.24	13.50	10.27	1.80	7.48	36.18	43.67	40.36
EVOLVINGNAV (ours)	63.22	48.83	35.43	12.98	19.31	61.32	86.18	70.15
	± 3.87	± 2.43	± 2.16	± 0.92	± 0.68	± 3.235	± 2.07	± 1.74

The groups mirror the source of each method’s advantage rather than suggesting that benchmark scores are interchangeable. EVOLVINGNAV leads every reported headline metric. On FindingDory, it reaches 63.22 HL-SR and 48.83 HL-SPL, improving over the strongest competing result by 10.78 and 25.58 points. On GOAT-Bench, its 35.43 SR, 12.98 SPL, and 19.31 Repeat-SR exceed the best baseline by 5.79, 7.53, and 2.94 points, respectively, indicating that the same memory interface transfers beyond explicitly evolving scenes. The advantage is most diagnostic on P4D-BENCH: First-Go SR rises to 61.32, 15.99 points above DynaMem, while Search SR and SPL reach 86.18 and 70.15. DynaMem remains strong after additional search (81.00 Search SR and 66.83 SPL), but its lower First-Go SR suggests that updating memory alone is insufficient to select the current destination before new evidence arrives. Released navigation systems are run through their native policies when possible; memory and prediction methods are inserted into a shared controller as specified in Section 5.3. Appendix tables retain the full FindingDory high-/low-level breakdown and disaggregate GOAT category, language, and image goals.

6.2 PREDICTIVE NAVIGATION ON P4D-BENCH

First-Go SR isolates whether a better current-state belief becomes a better physical destination. Search SR and cost then test whether probability mass beyond Top-1 is useful. Recovery SR is conditioned on an incorrect first prediction so that easy first-attempt successes do not mask poor evidence use. The relevant estimands are paired episode-level differences with confidence intervals that retain the pairing.

The complete N1–N3 comparison, including inspection, distance, and recovery columns, is provided in Appendix Table 8. Keeping the detailed protocol table with the supplementary diagnostics leaves this section focused on the estimands and qualitative behavior while preserving the full evaluation record.

Qualitative case study. Figure 4 traces one routine-world episode from stale memory to physical verification. The target bowl was last observed in a cabinet 14.77 hours before the query, but a dinner-cleanup transition relocates it to the kitchen sink. The predictive belief ranks the sink first, allowing P4D-NAV to verify the target in one inspection; the comparison routes expose how an obsolete last-seen state or a geometry-only ordering increases search cost.

Temporal and budget dependence. Memory staleness, inspection budget, and mobility regime define complementary axes of analysis. SR as a function of staleness tests robustness to observation gaps; SR as a function of inspection budget tests the value of alternatives beyond Top-1; and

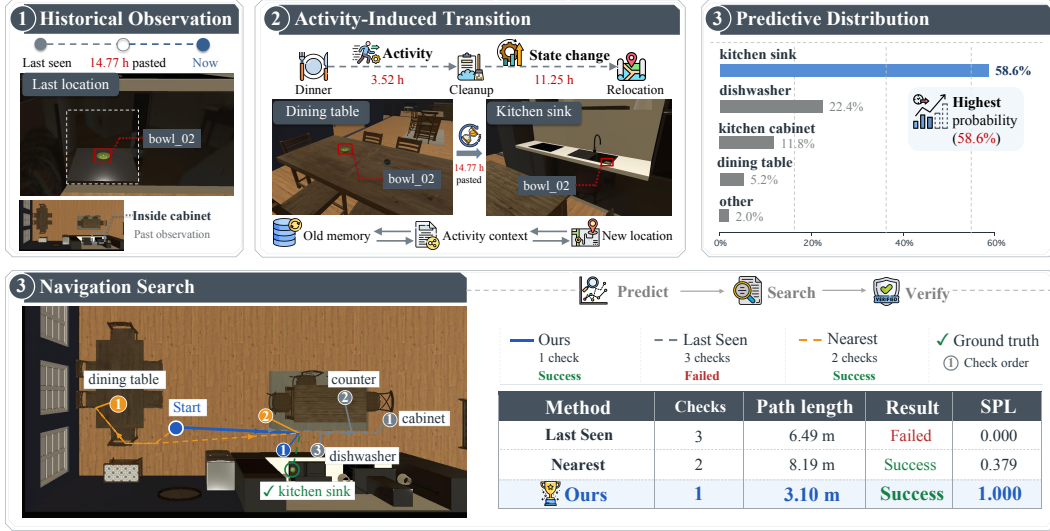


Figure 4: P4D-Bench case study comparing predictive, last-seen, and nearest-first search.

Table 4: Cross-environment navigation on one canonical task per simulator.

Method	Standard Simulation Worlds				Photorealistic Simulation				Evolving Simulation Worlds	
	R2R-CE		HM3D-OVON		PointNav		SAGE-Bench		P4D-BENCH	
	SR	SPL	SR	SPL	SR	SPL	SR	SPL	SR	SPL
<i>Navigation-trained or task-adapted</i>										
MTU3D [†] (Zhu et al., 2025)	-	-	-	-	-	-	-	-	-	-
SoftNav (Wu et al., 2026)	-	-	-	-	-	-	-	-	-	-
DUET [†] (Chen et al., 2022)	-	-	-	-	-	-	-	-	-	-
Uni-NaVid [†] (Zhang et al., 2025)	-	-	-	-	-	-	-	-	-	-
NaVILA [†] (Cheng et al., 2025)	-	-	-	-	-	-	-	-	-	-
D3D-VLP (Wang et al., 2026b)	-	-	-	-	-	-	-	-	-	-
<i>Zero-shot navigation without target-benchmark policy training</i>										
VLFM [†] (Yokoyama et al., 2024a)	-	-	-	-	-	-	-	-	-	-
AO-Planner [†] (Chen et al., 2025)	-	-	-	-	-	-	-	-	-	-
TANGO [†] (Ziliotto et al., 2025)	-	-	-	-	-	-	-	-	-	-
MSGNav [†] (Huang et al., 2026)	-	-	-	-	-	-	-	-	-	-
Uni-LaViRA [†] (Ding et al., 2026)	-	-	-	-	-	-	-	-	-	-
EVOLVINGNAV (ours)	-	-	-	-	-	-	-	-	-	-

paired routine gains separate learnable temporal structure from matched random transitions. These diagnostics require episode-level measurements and are not inferred from aggregate SR. Appendix Figure 7 visualizes these complementary staleness, budget, and mobility diagnostics.

6.3 GENERALIZATION ACROSS SIMULATION FIDELITY

Table 4 assigns one established task to each simulator rather than pooling incompatible goal definitions: R2R-CE on MP3D, HM3D-OVON on HM3D, PointNav on Habitat-GS, SAGE-Bench on InteriorGS, and P4D-BENCH on HSSD (Krantz et al., 2020; Yokoyama et al., 2024b; Xia et al., 2026; Miao et al., 2025). The first group uses navigation trajectories, task-specific fine-tuning, or reinforcement learning. The second performs zero-shot evaluation without training a navigation policy on the target benchmark. This distinction is important for EVOLVINGNAV, whose controller and foundation model remain frozen even though its world-state predictor is trained on causal histories.

We include methods only when the exact benchmark has published results or a released implementation supports a matched rerun; [†] marks the latter. Published values are transferred only when the split, sensor, action, and success protocols match. Otherwise, we rerun the released checkpoint and report the result as a reproduction rather than silently mixing protocols. Comparisons are therefore made vertically within each benchmark, not horizontally across tasks. Beyond average SR, the mesh-to-Gaussian and static-to-evolving changes reveal whether failures arise from visual transfer, goal grounding, or stale world state.

6.4 WHAT COMPONENTS PRODUCE THE GAIN?

Table 5: Module-level ablation of predictive belief and agent components.

Variant	Top-1	NLL	SR	SPL	Inspect.
w/o predictive belief	–	–	–	–	–
w/o 4D memory (latest observation)	–	–	–	–	–
w/o continuous-time encoding	–	–	–	–	–
w/o persistence–relocation head	–	–	–	–	–
w/o visibility calibration	–	–	–	–	–
w/o Bayesian evidence update	–	–	–	–	–
Full EVOLVINGNAV	–	–	–	–	–

The two top-level removals test the paper’s central system claims: whether prediction adds value beyond retrospective memory, and whether temporally versioned evidence adds value beyond the latest observation. The remaining rows localize gains within the predictor and agent loop. Joint interpretation of NLL and navigation cost distinguishes improved classification from improved decisions. In particular, the no-update and no-visibility variants separate the initial prediction from the agent’s use of later physical evidence. All removals preserve perception, candidates, and low-level control.

Table 6: Prediction-only results on temporal changed-only ($n = 322$) and LOSO change-balanced queries ($n = 644$). Ours reports mean $\pm$ standard deviation over five seeds; Top-1 is in percent, while MRR and NLL are unscaled.

Predictor	Temporal Top-1 $\uparrow$	MRR $\uparrow$	NLL $\downarrow$	LOSO Top-1 $\uparrow$	MRR $\uparrow$	NLL $\downarrow$
Uniform Random	9.32	0.2607	2.4509	11.96	0.3050	2.2980
Last Seen	0.00	0.1699	3.4011	50.00	0.5850	1.8825
Object Frequency	30.75	0.5476	1.7397	50.47	0.6441	1.8455
Time Frequency	33.85	0.5462	1.8400	49.53	0.6166	1.6455
Markov	18.94	0.4435	2.3772	49.38	0.6286	1.6982
Activity-Markov	25.16	0.4582	2.2436	49.84	0.6221	1.6183
Random Forest	20.50	0.4851	1.9181	58.07	0.7283	1.1441
Packed MLP	29.50	0.5225	2.1036	58.07	0.7173	1.3105
GRU	32.61	0.5471	1.8279	54.04	0.7016	1.2558
Original gated Transformer	34.16	0.5586	1.7292	51.71	0.6884	1.3157
Shared semantic prior	36.02	0.5957	1.5766	59.47	0.7462	1.0448
Ours	38.24 ± 1.42	0.5971 ± 0.0291	1.4925 ± 0.0867	61.18 ± 4.53	0.7523 ± 0.0176	0.9778 ± 0.0228

Current-state belief prediction. Before coupling prediction with navigation, we first isolate whether long-term memory can recover the current object state from stale observations. Table 6 evaluates two complementary settings. The temporal split contains changed-only queries and therefore removes trivial cases in which the last-seen state remains correct, while the LOSO split uses change-balanced queries to additionally evaluate transfer to held-out scenes. The absolute Top-1 values across the two settings are thus not directly comparable.

On temporal changed-only queries, Ours is best across all three metrics, reaching $38.24 \pm 1.42\%$ Top-1, 0.5971 ± 0.0291 MRR, and 1.4925 ± 0.0867 NLL. Relative to the strongest shared semantic prior, this is a 2.22-point Top-1 gain while also reducing NLL from 1.5766 to 1.4925. The consistent improvement in both argmax accuracy and distributional quality is important because the downstream agent uses the full location belief to order physical inspections.

Under LOSO evaluation, Ours again achieves the best result across all three metrics, reaching $61.18 \pm 4.53\%$ Top-1, 0.7523 ± 0.0176 MRR, and 0.9778 ± 0.0228 NLL. Compared with the strongest shared semantic prior, Ours improves Top-1 by 1.71 percentage points and reduces NLL from 1.0448 to 0.9778. These results suggest that structured long-term memory contributes information beyond shared semantic regularities and improves the transfer of current-state beliefs to unseen scenes.

Figure 5 complements the ranking metrics with probability quality. The structured prior follows empirical accuracy more closely than the direct Transformer, while evidence updating maintains this reliability and concentrates more predictions in the high-confidence bins. Thus, the update mechanism improves an action-facing belief rather than merely sharpening an uncalibrated score, which is essential when confidence determines the next inspection target.

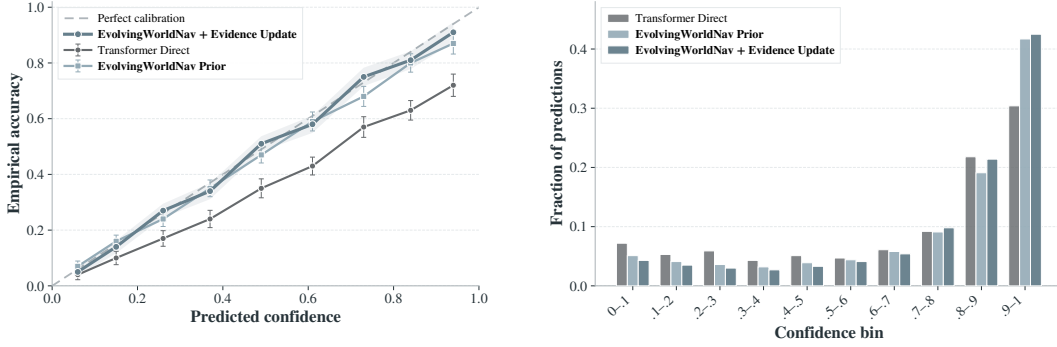


Figure 5: Calibration of current-state beliefs before and after evidence updates.

Because this table evaluates prediction independently of embodied execution, it does not by itself establish a navigation advantage. We therefore next evaluate whether these improved beliefs translate into more efficient search and navigation. Full predictor configurations, matched-input controls, and additional diagnostics are reported in Appendix C.2.

6.5 FOUNDATION-MODEL ADAPTATION

The full paired backbone matrix is moved to Appendix Table 9.

This experiment isolates the contribution of predictive memory from foundation-model capacity. Following embodied evaluations that span proprietary and open-weight VLMs (Qiao et al., 2025), we select two Qwen2.5-VL scales (Bai et al., 2025), two OpenAI controllers (OpenAI, 2026a;b), and two Gemini controllers (Google, 2026a;b). For every controller, the last-seen baseline and EVOLVINGNAV run the same episodes with identical perception, candidate locations, tools, prompts, action budgets, and low-level navigator. The baseline replaces predictive 4D belief with the last supported target state; EVOLVINGNAV keeps the VLM fixed and changes only the memory, prediction, and evidence-update pathway.

The decisive statistic is therefore the paired within-controller gain rather than the absolute ranking of VLMs. We report mean $\pm$ standard deviation over five seeds and compute Δ Search SR between matched configurations. Consistently positive gains across open-weight, high-capacity, and cost-oriented controllers would show that the navigation advantage persists across backbone families instead of arising from a particular proprietary VLM. Together with the VLM-independent prediction results in Table 6, this separates the quality of the world-state belief from the general reasoning strength of the controller.

6.6 REAL-WORLD EVALUATION

All numerical results in this subsection are measured from completed physical robot trials.

Predictive memory improves budgeted real-world search. We compare four methods on the primary LYNX M20 platform using 64 matched search blocks per method, equally divided between indoor and outdoor environments. Each run permits three inspections and 360 seconds. First-Go SR requires autonomous target confirmation at the first inspection; Search SR measures confirmation within the full episode budget.

Table 7: Primary LYNX M20 search results (64 episodes per method).

Method	First-Go SR $\uparrow$	Search SR $\uparrow$	Recovery SR $\uparrow$	Dist. (m)	Time (s)	Completed inspect.
Last Seen + Search	17.2	26.6	12.8	56.2	284	2.19
Time Frequency	21.9	31.3	14.0	52.9	277	2.05
Retrieval + Reasoning	26.6	37.5	17.1	50.3	281	2.03
EVOLVINGNAV	34.4	48.4	24.3	43.8	248	1.84

As shown in Table 7, EVOLVINGNAV raises Search SR from 37.5% for Retrieval + Reasoning to 48.4% and First-Go SR from 26.6% to 34.4%, while reducing mean travel from 50.3 m to 43.8 m. Recovery remains limited at 24.3%, so predictive belief does not eliminate end-to-end

failures. Indoor/outdoor, temporal-condition, recovery, and cross-platform breakdowns are reported in Appendix D.2.

7 CONCLUSION

We introduced persistent navigation in evolving worlds, a setting in which an embodied agent must act on a long-term but potentially stale account of its environment. EVOLVINGNAV provides a memory-belief-action loop for inferring task-relevant current state, while its P4D-NAV core operationalizes this problem through dynamic object goals: it separates persistence from relocation, represents variable scene-specific candidates, converts visibility-qualified observations into posterior evidence, and uses the evolving distribution to navigate, explore, and replan. P4D-BENCH grounds this question in human-trace-driven HSSD histories and paired static, routine, and random worlds spanning prediction, navigation, and embodied reasoning. The resulting formulation reframes navigation from executing against a remembered snapshot to acting on the world state best supported by accumulated and newly collected evidence. Quantitative conclusions will be added only after the multi-scene navigation evaluation is complete.

8 LIMITATIONS

Although EVOLVINGNAV is motivated by general task-relevant world evolution, the current predictive model is operationalized primarily through discrete portable-object states; structural reconfiguration and changing traversability require richer state and action models. P4D-NAV also assumes reliable entity association and an available set of candidate receptacles and inspection viewpoints; pose drift, missed detections, identity switches, or missing candidates can corrupt both prediction and navigation. Its discrete belief represents arbitrary continuous placements less directly than continuous generative models. The source datasets are complementary rather than jointly observed, so P4D-BENCH episodes are human-trace-grounded simulations, not reconstructions of one household. The main protocol freezes the target after the query; N4 is required to test simultaneous target and agent motion. Finally, visibility-aware updates reduce weak negative evidence but cannot guarantee recovery when beliefs and perception fail together or routines shift.

The real-world evaluation covers indoor and outdoor environments and three physical robot platforms, but does not represent unrestricted long-term deployment. Targets may change between observation sessions but remain stationary during each individual search episode; continuously moving targets would additionally require online tracking and motion prediction. Moreover, all three evaluated platforms are from the same manufacturer, and the current study does not establish transfer to substantially different sensing, localization, or control stacks. Finally, real-world performance still depends on the underlying perception and navigation systems: localization errors, missed detections, and navigation failures can prevent task completion even when the high-level belief is correct.

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A ADDITIONAL METHOD DETAILS

This section gives the formal episode interface, memory representation, baseline specification, and implementation details that support the method described in the main paper.

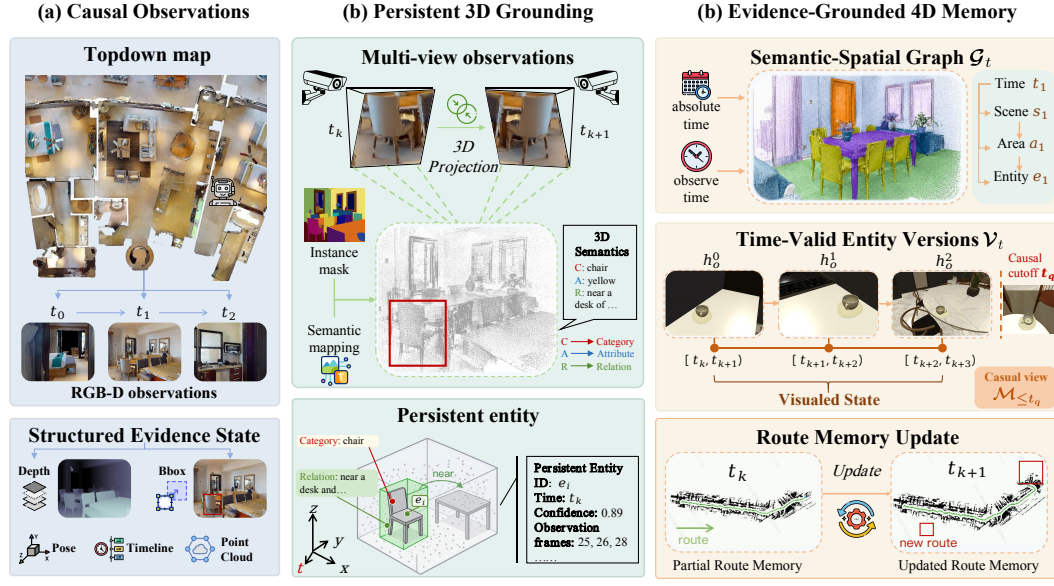


Figure 6: Temporally versioned 4D memory construction.

A.1 FORMAL EPISODE INTERFACE

Each navigation episode references one causal prediction query. Public fields include schema version, episode and query identifiers, split, task type, scene and household identifiers, world variant, target description, query time, agent start pose, candidate-state identifiers, budgets, and success criteria. Evaluation-private fields include the current state, target pose, valid goal viewpoints, hidden transition class, oracle path, mobility label, and query-time activity. Train and validation releases may expose analysis labels in metadata; the formal test release replaces them with null values and computes grouped metrics in a private evaluator.

The high-level track exposes

NAVIGATE_TO(state), INSPECT(state), STOP, NOT_FOUND,

with a fixed navigation backend. The low-level track exposes Habitat-style forward, turn, look, and stop actions. The main paper uses the high-level track to isolate predictive state selection; low-level results measure sensitivity to control and perception.

A.2 DETAILED 4D MEMORY CONSTRUCTION

Figure 6 illustrates how observations are converted into persistent entity versions. Each observation contributes spatial evidence and temporal validity, allowing the memory to preserve both previous and current states. For image coordinate $\tilde{\mathbf{u}} = [u, v, 1]^\top$ with depth $D_t(u, v)$, the corresponding world point is

$$\mathbf{x}^w = \text{proj}_3 \left(\mathbf{T}_{w \leftarrow c, t} \begin{bmatrix} D_t(u, v) \mathbf{K}^{-1} \tilde{\mathbf{u}} \\ 1 \end{bmatrix} \right), \quad (14)$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\mathbf{T}_{w \leftarrow c, t}$ is the camera-to-world transform. Cross-view association uses semantic compatibility, 3D overlap, temporal consistency, and source confidence. The m -th version of entity o is

$$h_o^m = ([t_{o,s}^m, t_{o,e}^m], s_o^m, c_o^m, f_o^m, \mathcal{P}_o^m), \quad (15)$$

containing a validity interval, candidate state, confidence, visual feature, and evidence handles. A state change appends a version without erasing the earlier trajectory, preserving both causal history and observation provenance.

A.3 FULL BASELINE SPECIFICATION

Last Seen assigns all mass to the latest positive state. Frequency Prior estimates $P(s \mid o)$ from training data. Markov Transition estimates $P(s_{k+1} \mid s_k, o)$; Time-Conditioned Prior additionally

conditions on public hour and weekday bins. Instance Hotspot uses only the target instance’s training trajectory. GRU Direct and Transformer Direct receive the same event tokens and candidate encoder as P4D-NAV but directly normalize candidate logits without the persistence factorization. P4D-Belief Prior uses Equation (9) once at episode start. Full P4D-NAV applies Equation (12) and Equation (10) after each inspection.

The Oracle Activity diagnostic may use the hidden activity and source location but is excluded from fair rankings. Oracle Current State receives the private target state and measures remaining perception and navigation error. External predictive methods are adapted only through their public inputs; any use of privileged simulator state must be identified and excluded from the main comparison.

A.4 IMPLEMENTATION DETAILS

Unless otherwise noted, the history encoder has three Transformer layers, hidden width 128, four attention heads, and at most 64 events. The predictive model contains approximately –million trainable parameters and is optimized with AdamW using validation NLL for checkpoint selection. Candidate encoders are shared across scenes. The main high-level track fixes candidate viewpoints and low-level planning; a Habitat low-level action track evaluates the complete embodied stack. Final learning rate, batch size, epoch count, hardware, simulator version, detector, visibility thresholds, and repeated-run protocol will be frozen before test evaluation.

B BENCHMARK CONSTRUCTION AND EVALUATION PROTOCOL

This section documents episode validity, leakage controls, mobility assignment, and the metrics used for all reported comparisons.

B.1 EPISODE VALIDITY AND LEAKAGE AUDIT

An episode is retained only if: (i) the target has at least one positive pre-query observation; (ii) the target state is known to the private evaluator and physically instantiated without collision; (iii) at least one valid viewpoint lies on the NavMesh; (iv) the start is on the NavMesh, at least 3 meters from the target, and does not reveal it; and (v) all public histories terminate at or before the query. A successful stop must be within 1 meter geodesic distance of a valid viewpoint, observe the target above a frozen visible-fraction threshold, and identify the correct instance or category.

We audit serialized examples for future timestamps, hidden event fields, filenames or indices correlated with answers, duplicate event groups across splits, and inconsistent paired variants. Static, routine, and random members of a pair must have identical public query fields. Invalid simulator episodes are reported separately and are not counted as ordinary failures.

B.2 MOBILITY-TYPE ASSIGNMENT

The frozen rule is

$$\text{type}(o) = \begin{cases} \text{stable}, & r_{\text{move}}(o) < \tau_{\text{move}}, \\ \text{activity}, & G_{\text{act}}(o) \geq \tau_{\text{act}}, \\ \text{personal}, & C_{\text{habit}}(o) \geq \tau_{\text{habit}}, \\ \text{irregular}, & \text{otherwise.} \end{cases} \quad (16)$$

All thresholds are selected on training data before navigation evaluation. When the generator specifies a mobility profile, the same statistics are used to verify that realized trajectories exhibit the intended behavior. Every group must cover several categories, instances, and scenes, and no category may uniquely reveal one mobility type.

B.3 METRIC DEFINITIONS

Let S_i be episode success, L_i the executed path, and L_i^* the oracle shortest path to the query-time goal under the same start. Standard SPL is

$$\text{SPL} = \frac{1}{N} \sum_{i=1}^N S_i \frac{L_i^*}{\max(L_i, L_i^*)}. \quad (17)$$

First-Go SR counts success when only the initial predicted candidate may be inspected. Recovery SR conditions on an incorrect first candidate and measures whether the target is subsequently found. Excess path is $L_i - L_i^*$ for static-after-query episodes. For online-dynamic episodes, the oracle is trajectory aware and may intercept the target; we therefore report both success within the common budget and $L_i - L_{i,\text{dyn}}^*$.

Calibration metrics operate on the full candidate belief before navigation and after each update. ECE bins maximum confidence, Brier score evaluates all candidates, and true-state rank avoids hiding useful changes below Top-1. Search-cost curves plot success against allowed inspections and travelled distance rather than reporting one arbitrary budget.

C ADDITIONAL EXPERIMENTAL RESULTS

This section provides extended quantitative results omitted from the main paper for space constraints.

C.1 P4D-BENCH DETAILED RESULTS

Table 8 expands the routine-world navigation comparison with inspection count, travel distance, and recovery success; lower is better for the two cost columns. Table 9 tests backbone robustness under matched tools, prompts, episodes, and action budgets.

Table 8: Detailed P4D-Bench navigation results on routine worlds.

Method	First-Go SR ↑	Search SR ↑	SPL ↑	Inspect. ↓	Distance ↓	Recovery SR ↑
<i>Navigation and persistent memory</i>						
VLFM (Yokoyama et al., 2024a)	—	—	—	—	—	—
FindingDory Agent (Yadav et al., 2025)	—	—	—	—	—	—
LagMemo GLUE (Zhou et al., 2025)	—	—	—	—	—	—
HOV-SG (Werby et al., 2024)	—	—	—	—	—	—
DynaMem (Liu et al., 2024)	—	—	—	—	—	—
<i>Predictive world and state modeling</i>						
SLaTe-PRO (Patel et al., 2023)	—	—	—	—	—	—
SGM+NEP (Kurenkov et al., 2023)	—	—	—	—	—	—
FlowMaps (Argenziano et al., 2026)	—	—	—	—	—	—
Perpetua (Saavedra-Ruiz et al., 2026)	—	—	—	—	—	—
P4D-NAV (ours)	—	—	—	—	—	—
Oracle Current State	—	—	—	—	—	—

Table 9: Backbone robustness on P4D-Bench under matched evaluation conditions.

Family	Fixed controller endpoint	Last-seen Agent			EVOLVINGNAV			Δ Search SR ↑
		First-Go SR ↑	Search SR ↑	SPL ↑	First-Go SR ↑	Search SR ↑	SPL ↑	
Qwen	Qwen2.5-VL-7B-Instruct	—	—	—	—	—	—	—
	Qwen2.5-VL-72B-Instruct	—	—	—	—	—	—	—
OpenAI	GPT-5.5 / gpt-5.5-2026-04-23	—	—	—	—	—	—	—
	GPT-5.6 Luna / gpt-5.6-luna	—	—	—	—	—	—	—
Google	Gemini 3.1 Pro / gemini-3.1-pro-preview	—	—	—	—	—	—	—
	Gemini 3.5 Flash / gemini-3.5-flash	—	—	—	—	—	—	—
Average gain								—

C.2 CROSS-BENCHMARK RESULTS

The native FindingDory control-track metrics are expanded in Table 10, while Table 11 reports GOAT-Bench results by goal modality.

Table 10: FindingDory results by control track.

Method	HL-SR	HL-SPL	LL-SR	LL-SPL
FindingDory Agent	—	—	—	—
LagMemo GLUE	—	—	—	—
HOV-SG	—	—	—	—
DynaMem	—	—	—	—
EVOLVINGNAV	—	—	—	—

Table 11: GOAT-Bench results by goal modality.

Method	Category goal		Language goal		Image goal	
	SR	SPL	SR	SPL	SR	SPL
VLFM	—	—	—	—	—	—
ZSON	—	—	—	—	—	—
FindingDory Agent	—	—	—	—	—	—
LagMemo GLUE	—	—	—	—	—	—
EVOLVINGNAV	—	—	—	—	—	—

C.3 EVOLUTION-AWARE DIAGNOSTICS

Table 12 breaks search success down by hidden mobility type, and Table 13 compares paired static, routine, and random worlds. These matched variants isolate evolution from scene-frequency shortcuts.

Table 12: Search SR by hidden mobility type.

Method	Stable	Activity routine	Personal habit	Irregular	Macro
Last Seen	—	—	—	—	—
Time-Conditioned	—	—	—	—	—
Instance Hotspot	—	—	—	—	—
Transformer Direct	—	—	—	—	—
P4D-NAV	—	—	—	—	—

Table 13: Paired-world navigation results.

Method	Static SR	Routine SR	Random SR	Routine gain over Last Seen
Last Seen	—	—	—	—
Markov Transition	—	—	—	—
Transformer Direct	—	—	—	—
P4D-NAV	—	—	—	—

Effect of memory staleness and inspection budget. Figure 7 complements these aggregate tables by showing how performance changes with temporal staleness, inspection budget, and hidden mobility type. These episode-level diagnostics are not inferred from aggregate success rates.

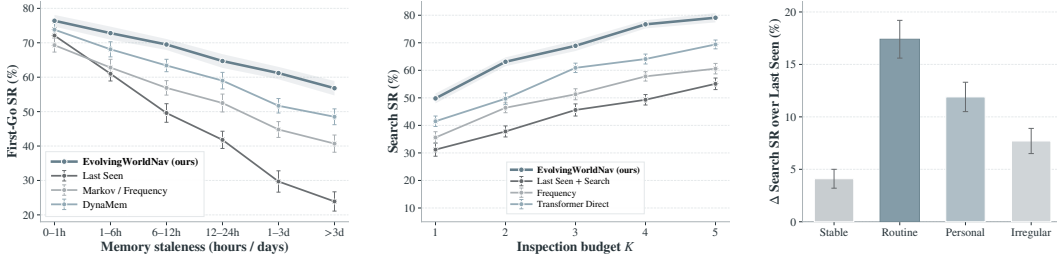


Figure 7: Predictive diagnostics under memory staleness and search budgets.

D REAL-WORLD EVALUATION

This section reports the physical evaluation, first describing the robot platforms and then disaggregating the measured M20 results by environment, temporal condition, and qualitative case.

D.1 ROBOT PLATFORMS

We conduct the real-world evaluation using three commercial robotic platforms from DEEP Robotics: LYNX M20, X30, and Lite3. Table 14 summarizes the platforms used in the physical evaluation. Platform-specific perception and low-level navigation are abstracted behind a common interface, while the EVOLVINGNAV memory, prediction, belief-update, and high-level search components remain unchanged.

The reported endurance and range values are manufacturer laboratory specifications and may differ from those obtained under the indoor and outdoor conditions used in our evaluation. We therefore do not use manufacturer-rated mobility or endurance as experimental outcomes. Absolute completion

Table 14: Robot platforms used in real-world evaluation.

Platform	Morphology	Standing size	Mass	Endurance / range
LYNX M20	Wheel-legged	820 × 430 × 570 mm	~35 kg	3 h / 15 km unloaded
X30	Quadruped	1000 × 695 × 470 mm	56 kg	2.5–4 h / ≥10 km
Lite3 (LiDAR)	Quadruped	610 × 370 × 496 mm	13.5 kg	1.5–2 h / 2.7 km

time is compared only between methods executed on the same robot platform. The M20 uses dual 96-line LiDARs and wide-angle cameras; X30 provides integrated perception and autonomous navigation; and the Lite3 LiDAR configuration exposes obstacle avoidance and autonomous navigation through its platform interface.

D.2 ENVIRONMENT BREAKDOWN

Table 15 separates indoor and outdoor trials. The outdoor setting has longer routes and lower success for all methods, while the direction of the EVOLVINGNAV gain is consistent across both environments.

Table 15: M20 results by environment.

Method	Indoor			Outdoor		
	First-Go	Search SR	Dist.	First-Go	Search SR	Dist.
Last Seen + Search	18.8	31.3	28.6	15.6	21.9	83.8
Time Frequency	25.0	34.4	27.2	18.8	28.1	78.6
Retrieval + Reasoning	31.3	40.6	25.1	21.9	34.4	75.5
EVOLVINGNAV	40.6	53.1	22.4	28.1	43.8	65.2

D.3 TEMPORAL CONDITION ANALYSIS

Table 16 shows that predictive memory is most useful when recurring changes provide a usable historical signal. Last Seen remains strong when nothing changes, whereas all methods degrade as regularity weakens.

Table 16: M20 Search SR (%) by temporal condition.

Method	Unchanged	Routine-consistent	Weak-routine	Broken-routine
Last Seen + Search	68.8	18.8	12.5	6.3
Time Frequency	50.0	50.0	18.8	6.3
Retrieval + Reasoning	62.5	43.8	25.0	18.8
EVOLVINGNAV	62.5	68.8	37.5	25.0

Recovery and cross-platform transfer. Among the 37 M20 episodes in which EVOLVINGNAV completed an unsuccessful first inspection, nine later succeeded, giving the 24.3% conditional Recovery SR in the main table. The matched no-update ablation is required to isolate Bayesian updating from additional search. Table 17 further shows the same direction of improvement on a 16-block subset executed by all three platforms; the small sample does not establish equal performance across robots.

Table 17: Cross-platform transfer results.

Platform	Method	First-Go SR	Search SR	Dist. (m)
LYNX M20	Last Seen + Search	18.8	25.0	55.4
	EVOLVINGNAV	37.5	50.0	44.6
X30	Last Seen + Search	18.8	25.0	54.1
	EVOLVINGNAV	31.3	43.8	46.8
Lite3	Last Seen + Search	18.8	31.3	51.8
	EVOLVINGNAV	25.0	43.8	45.9

D.4 QUALITATIVE CASES

Figures 8 and 9 show the two physical task sequences. The indoor sequence covers person verification, document collection, and navigation to a meeting room; the outdoor sequence shows obstacle-triggered replanning while searching for shared bikes. The figures illustrate execution traces and do not add aggregate success-rate measurements.



Figure 8: Indoor real-world execution case.

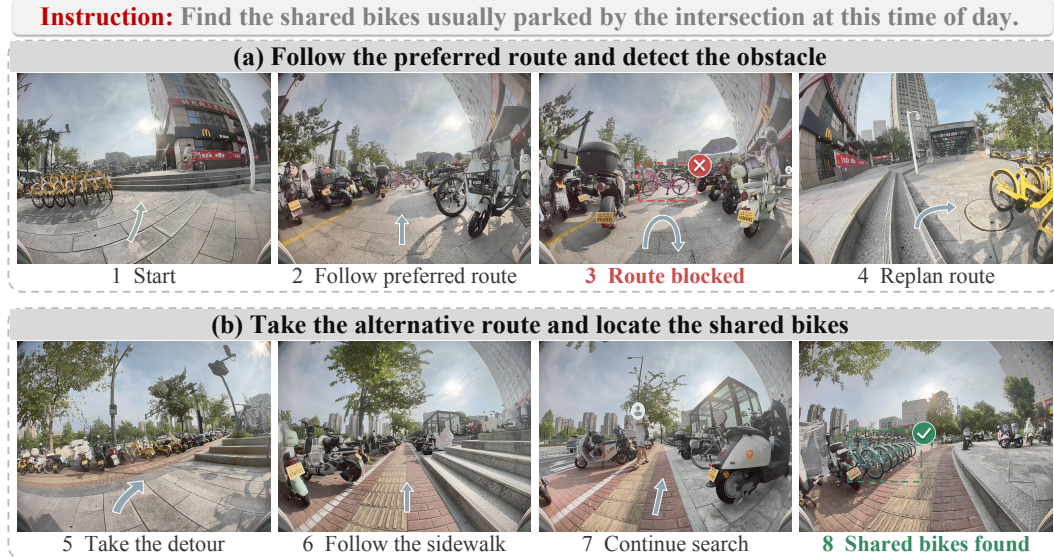


Figure 9: Outdoor real-world execution case.

E REPRODUCIBILITY AND RESPONSIBLE AI

E.1 REPRODUCIBILITY CHECKLIST

The completed release will report:

- source dataset versions, licenses, mappings, filters, and canonical-record statistics;
- HSSD and Habitat versions, scene identifiers, object assets, placement checks, and NavMesh settings;
- split manifests, paired-example identifiers, checksums, and leakage-audit output;
- model width, layers, attention heads, time features, history length, optimizer, learning rate, epochs, seeds, and checkpoint rule;
- detector and depth inputs, visibility computation, candidate and action budgets, evidence likelihoods, and planner hyperparameters;
- hardware, wall-clock latency, memory growth, repeated-run aggregation, and confidence intervals; and
- reproduced versus quoted baseline values, grouped results, failure labels, and all component-aligned ablations.

E.2 ETHICS

Persistent visual memory may capture people or private spaces, while an incorrect belief may induce unnecessary motion or unsafe behavior. Deployments should obtain consent, enforce retention and access controls, preserve provenance, and retain platform-specific collision avoidance and emergency stops outside the learned model. Our benchmark compiles existing records into simulated HSSD episodes and introduces no new human-subject study.

E.3 AI USE

Generative AI tools assisted manuscript editing, LaTeX checks, and local engineering. They did not generate benchmark observations, ground-truth labels, or reported scores. The authors reviewed all assisted material and remain responsible for the content and artifacts.